

Impact of Nutritional Information on Food Choices

Evidence from a Classroom Randomized Controlled Trial

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Outline

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- 3 Results
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Research Question

Main Research Question

Does exposure to nutritional information influence food choices?

Motivation

- Poor dietary habits contribute to major public health challenges.
- Information campaigns are widely used to improve nutrition outcomes.
- Understanding whether information changes behaviour is important for policy design.

Hypothesis

Null Hypothesis (H_0)

There is no difference in the likelihood of choosing healthier food options between participants exposed to nutritional information and those who were not exposed.

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Alternative Hypothesis (H_1)

Exposure to nutritional information increases the likelihood of choosing healthier food options.

Conceptual Framework

Treatment
(Nutritional Information)



Knowledge and Awareness



Food Choice Behaviour



Healthy Food Selection

Experimental Design

- Classroom-based Randomized Controlled Trial (RCT)
- Treatment group received nutritional information
- Control group received no information
- Outcome variable: Healthy food choice
- Objective: Estimate the Average Treatment Effect (ATE)

Data and Randomization

- Participants: Learning Days 2026 attendees
- Sample size: $N = 19$
- Blocked randomization based on:
 - Gender
 - Language
- Ensures balanced treatment assignment across groups

Implementation Report

- Experiment conducted during classroom sessions
- Treatment group received nutritional information immediately before lunch
- Food choices observed after treatment

Limitations

- Small sample size
- Inconsistent treatment fidelity
- Classroom environment effects

Sample Characteristics

- Total observations: 19
- Four language-gender blocks
- Balanced treatment allocation
- Similar participant characteristics across groups
- Randomization helps ensure internal validity

Estimation Model

$$Y_i = \beta_0 + \beta_1 \textit{Treatment}_i + \beta_2 \textit{Block}_i + \varepsilon_i$$

Where:

- Y_i = Healthy food choice
- Treatment = Nutritional information
- Block = Language-Gender categories
- β_1 measures the treatment effect

Regression Results

Variable	Estimate	Std. Error	t-value	p-value
Intercept	0.8929	0.3140	2.843	0.0130*
Treatment	0.2143	0.1954	1.097	0.2913
F. French	-0.0357	0.3867	-0.092	0.9277
M. English	-0.1667	0.3446	-0.484	0.6361
M. French	-0.6250	0.3337	-1.873	0.0821.

Main Findings

- The treatment coefficient is positive (0.214), suggesting that nutritional information increased the probability of choosing a healthy food option by about 21 percentage points.
- However, the effect is not statistically significant ($p = 0.291$).
- Relative to the reference group (Female English speakers), only Male French participants show a sizeable negative difference, although significance is weak ($p = 0.082$).

Interpretation of Results

- The treatment coefficient is positive (0.214), indicating a higher likelihood of choosing healthy food among treated participants.
- The estimated increase is approximately 21 percentage points.
- However, the p-value (0.291) exceeds conventional significance levels.
- Therefore, we fail to reject the null hypothesis.
- The results suggest a positive but statistically inconclusive effect of nutritional information.

Why Was There No Effect?

- Information exposure was very brief.
- Food preferences may be difficult to change immediately.
- Small sample size reduced statistical power.
- Participants may already have possessed nutrition knowledge.
- Behavioural change often requires repeated interventions.

Policy Implications

- Information alone may not be sufficient to change behaviour.
- Behavioural change often requires multiple interventions.
- Nutrition programs may benefit from behavioural nudges.
- Context matters when designing public health interventions.

Recommendations

- Increase sample size in future studies.
- Improve treatment fidelity.
- Use more intensive and repeated information campaigns.
- Measure outcomes over a longer time horizon.
- Combine information with behavioural incentives.

Key Takeaways

- 1 Nutritional information alone did not significantly affect food choices.
- 2 No evidence of differences across language-gender groups.
- 3 Behavioural change may require stronger interventions.
- 4 Larger studies are needed to detect small treatment effects.

Conclusion

Main Message

A short nutritional information intervention did not significantly influence healthy food choices among participants in this classroom randomized controlled trial.

Thank You

Questions and Discussion